

Efflorescence in concrete roof tile

Generally, the occurrence of white spots on the surface of concrete products is called efflorescence.

In concrete roof tiles, during production processes can appear changes in tile surface, among them an uniformed whitening, various white dots with different diameter and irregular spots, all of these can be called **efflorescence**.

The **efflorescence** always prejudice the appearance and the color of the concrete tiles

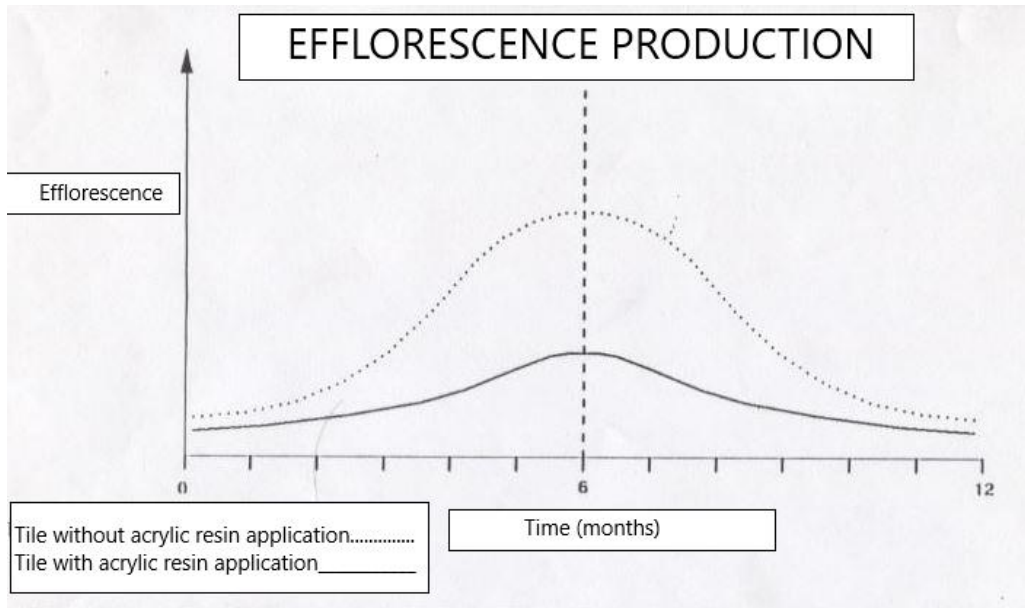
All these distinguishing marks show up mainly in the external surface of the tile and can be produced when the humidity dissolves the soluble compounds (sulphates, carbonates, silicates, chlorides, nitrates, etc) from concrete tile body and carry them to the surface where they react with the air and, in some extreme cases, by evaporating they leave a mineral deposit (crystalline salts), usually is Calcium Carbonate. CaCO_3

It is necessary that all the mixing components (sand, water, pigment) have the less quantity of calcium, independent of that provided by the cement itself, in order to avoid the creation of soluble compounds that can develop in **efflorescence**.

The main types of **efflorescence** are the following:

1. **Primary efflorescence** - an uniformed whitening of tile surface can be caused by an inappropriate curing cycle to the tiles (with curing chambers or tunnels) that can blanch the colour tonality and is identified at the exit of the curing chambers or tunnels.
2. **Condensing efflorescence** – a lot of small white dots on tile surface, that can correspond to cold or too warm areas and that are created during the curing process. This is identified at the exit of the curing chambers or tunnels.
3. **Frontal/rear efflorescence** with isolated spots on the surface. This efflorescence can be generated during cutting process due to the water spraying to clean the blades that can generate a humidity excess, that can be detected after the curing, at the exit of the curing chambers or tunnels.
4. **Efflorescence on upper & lower tile surfaces** that appear when the tiles are on the yard or when already installed on the roof, that appear with very intense white spots, Calcium Carbonate. This is the most extreme form of **efflorescence** with the process of carbonization (chemical reaction where the calcium hydroxide reacts with carbon dioxide and forms insoluble calcium carbonate form: $\text{Ca(OH)}_2 + \text{CO}_2 \rightarrow \text{CaCO}_3 + \text{H}_2\text{O}$). It appears just during some periods of the year with extreme weather conditions, like heavy rain, stagnant water, cold temperatures, strong changes heat / cold, condensation, snow, dew. The efflorescence varies a lot in its speed and shape depending on the humidity levels.

The possibility of having efflorescence on the tile surface is correlated to the application of the acrylic sealer before the tiles packing.



HOW TO PREVENT OR REDUCE EFFLORESCENCE ON THE TILES:

The first things that helps to avoid **efflorescence** is the usage of washed sands, with a good granulometry and Portland cement with low-grade of alkalies in order to have a mixing with low water absorption.

If, using the best raw materials possible, the problem with **efflorescence** persists on the upper or lower surfaces, we should try to minimize the effects at least on the upper surface that is the visible part on a roof.

TILE UPPER SURFACE:

In order to avoid **efflorescence** on that area is essential to apply an acrylic sealer on the entire upper/outer surface of the cured tile after depalletting and before packing.

Also wrapping the tiles packages helps to preserve the tiles surface form the formation of **efflorescence**.

TILE LOWER SURFACE:

In order to reduce **efflorescence** on lower surface of the tile it is important to avoid the water absorption in the concrete mix and this can be reached putting an additive (waterproofing) that can reduce/eliminate the penetration of water into the body of the tile, blocking the pores of the concrete in order to reduce that the soluble salts come out to the tile surface.

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